

THE COST OF INACTION: Exploring The Consequences Of Ignoring IIoT SECURITY RISKS

As industrial IoT (IIoT) integration surges, so do security concerns. Let's delve into the rising threat landscape and the role of the security model in fortifying IIoT defences and safeguarding critical infrastructure



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Industrial IoT (IIoT) security has gained tremendous importance in recent years. With the rise of IIoT and the increasing integration of operational technologies (OT) with the internet, the need for robust security measures has become paramount. Industrial IoT, often referred to as OT, has transformed industries across the board. The advent of 5G technology has made OT assets integral to the manufacturing, health-care, and robotics sectors. These technologies have improved operational efficiency and enhanced worker safety by reducing accidents in industrial settings.

In the past, OT systems operated in isolation; however, they are interconnected today, leading to more efficient and complex systems. Remote operations have also become increasingly common, highlighting the need for secure IIoT infrastructure. While IIoT offers numerous benefits, it also introduces security challenges. Connecting OT systems to the internet inherently involves some level of risk. Recent news reports have highlighted

industrial cyberattacks, such as the Ukrainian electric grid attack and the Saudi oil refinery attack, showcasing the vulnerabilities of increased connectivity.

Ransomware attacks, in particular, have emerged as a significant threat to the manufacturing sector. In 2021 alone, there were ten successful ransomware attacks targeting manufacturing companies. The consequences of such attacks are severe, disrupting production, encrypting data, and causing financial losses.

Challenges in India

India is not immune to these threats. Recent attacks on organisations like Oil India, Dr Reddy's Laboratories, and Tata Power have underscored the critical nature of IIoT security. The healthcare sector has not been spared, with institutions like AIIMS falling victim to ransomware attacks, paralysing essential services. One of the most concerning trends is the rise of state-sponsored cyberattacks. Countries like China and others are known to engage in targeted campaigns, launching attacks on critical infrastructure worldwide. These attacks have been a constant threat, with significant incidents reported yearly.

The connectivity dilemma with the expansion of IIoT has exposed security gaps. This is depicted in Fig. 1.

Many OT systems are externally accessible, with public-facing applications and internet accessibility, making them susceptible to attacks. Vulnerabilities in industrial control systems (ICS) have increased by 50%, with 61% being remotely exploitable.

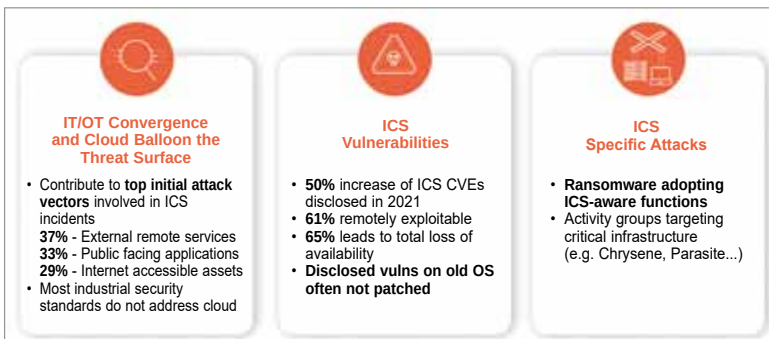


Fig. 1: Security gaps due to connectivity dilemma with the expansion of IIoT
(Credit: Palo Alto Networks)